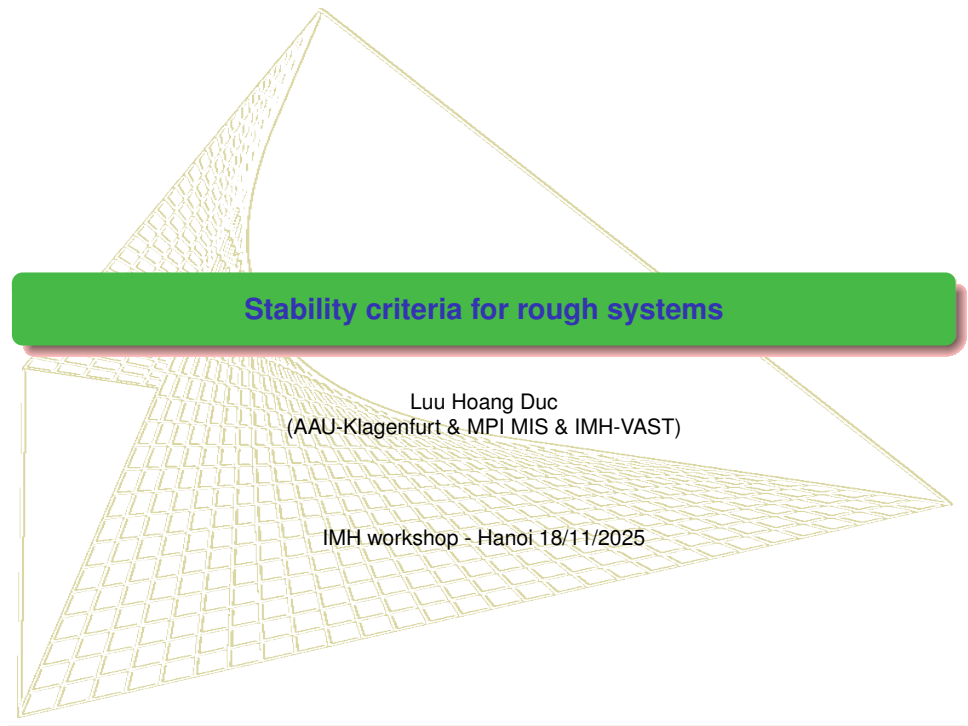




Stability criteria for rough systems

Luu Hoang Duc
(AAU-Klagenfurt & MPI MIS & IMH-VAST)

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joint work with Jürgen Jost

Stochastic stability

- Asymptotic stability for deterministic systems: Lyapunov (1892) $\epsilon - \delta$ language.
- Stochastic stability for Markov systems: Khasminskii (1979), Skorohod (1987), Molchanov, Arnold, Pardoux, Baxendale, Mao... and many others in control theory.

$$dy_t = f(y_t)dt + g(y_t)dB_t, \quad (1.1)$$

$f : \mathbb{R}^d \rightarrow \mathbb{R}^d, g : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$ glob. Lischitz continuous w.r.t. C_f, C_g and f strictly dissipative, i.e. $\langle y, f(y) \rangle \leq -d\|y\|^2$. By Itô's formula

$$d\mathbb{E}\|y_t\|^2 = \mathbb{E}\left(2\langle y_t, f(y_t) \rangle + \|g(y_t)\|^2\right)dt + \mathbb{E}2\langle y_t, g(y_t)dB_t \rangle \quad (1.2)$$

$$\leq -(2d - C_g^2)\mathbb{E}\|y_t\|^2 dt. \Rightarrow C_g^2 < 2d! \quad (1.3)$$

- Later generalize to Lyapunov function $V : \mathbb{R}^d \rightarrow \mathbb{R}_+$ and $\exists d_1, d_2 > 0$

$$\langle \nabla V(z), f(z) \rangle \leq d_1 - d_2 V(z), \quad \forall z \in \mathbb{R}^d.$$

- When B is replaced by B^H for $H \in (\frac{1}{3}, \frac{1}{2})$, one needs Lyons' rough path theory and more regularity, e.g. $(H_g) g$ is in $C_b^3(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^m, \mathbb{R}^d))$ with $C_g = \max\{\|D^{(i)}g\|_\infty : i = 0, \dots, 3\}$.
- Ergodicity theory by Mattingly, Hairer, Scheutzow, Meyn, Tweedie,... even for B^H .
- Related topics on RDS and random (numerical) attractors: Crauel, Kloeden, Arnold, Duan, Schmalfuss, Garrido-Atienza,... Even more for SPDEs with the semi-group technique.

Doss-Sussmann transformation

- Rough ODE $dy_t = f(y_t)dt + g(y_t)d\mathbf{x}_t$ is understood in the Gubinelli rough integral form

$$y_t = y_0 + \int_0^t f(y_s)ds + \int_0^t g(y_s)d\mathbf{x}_s. \quad (1.4)$$

- the solution $\phi(\cdot, \mathbf{x}, \phi_a)$ of $d\phi_u = g(\phi_u)d\mathbf{x}_u$, $u \in [a, b]$, $\phi_a \in \mathbb{R}^d$ is C^1 w.r.t. ϕ_a , and $\frac{\partial \phi}{\partial \phi_a}(\cdot, \mathbf{x}, \phi_a)$ is the solution of the linearized system

$$d\xi_u = Dg(\phi_u(\mathbf{x}, \phi_a))\xi_u d\mathbf{x}_u, \quad u \in [a, b], \xi_a = Id, \quad (1.5)$$

- Cass-Litterer-Lyons (2013): Stopping times

$$\tau_0 = a, \quad \tau_{i+1} := \inf \left\{ t > \tau_i : \|\mathbf{x}\|_{p\text{-var}, [\tau_i, t]} = \gamma \right\} \wedge b.$$

- The Doss-Sussmann transformation $y_t = \phi_t(\mathbf{x}, z_t) = z_t + \int_{\tau_i}^t g(\phi_s(\mathbf{x}, z_t))d\mathbf{x}_s$ on $[\tau_i, \tau_{i+1}]$

$$\dot{z}_t = \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} f(\phi_t(\mathbf{x}, z_t)) = (I + \psi_t)f(z_t + \eta_t), \quad t \in [\tau_i, \tau_{i+1}], \quad z_{\tau_i} = y_{\tau_i}. \quad (1.6)$$

$$\eta_t := y_t - z_t, \quad \text{and} \quad \psi_t := \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} - Id, \quad \forall t \in [\tau_i, \tau_{i+1}].$$

Lyapunov function

Classical condition on the gradient ∇V along the vector field f : $\exists d_1, d_2 > 0$

$$\langle \nabla V(z), f(z) \rangle \leq d_1 - d_2 V(z); \Rightarrow \langle \nabla V(z), (I + \psi)f(z + \eta) \rangle \leq d_1 - d_2 V(z) \quad \forall [\tau_i, \tau_{i+1}]? \quad (1.7)$$

But η_t, ψ_t varies as stochastically as x_t , e.g. $\eta_t = \int_{\tau_i}^t g(\phi_s(\mathbf{x}, z_t)) dB_s(\mathbf{x})$.

($\mathbf{H}_V$) There exists for a Lyapunov-type function V for the drift f that satisfies

- there exists functions $\alpha, \beta \in \mathcal{K}'_\infty$ such that

$$\alpha(\|z\|) \leq V(z) \leq \beta(\|z\|), \quad \forall z \in \mathbb{R}^d; \quad (1.8)$$

- there exists a constant $L_V > 0$ such that

$$\|\nabla V(z)\| \leq L_V, \quad \forall z \in \mathbb{R}^d; \quad (1.9)$$

- there exists a sufficiently small constant $\lambda \in (0, 1)$ and constants $C_\lambda, \delta > 0$ such that

$$\sup_{\|\psi\|, \|\eta\| \leq \lambda} \langle \nabla V(z), (I + \psi)f(z + \eta) \rangle \leq C_\lambda - \delta V(z), \quad \forall z \in \mathbb{R}^d. \quad (1.10)$$

① (1.10) is satisfied for a range $0 \leq \lambda < \lambda_0$ with C_{λ_0}, δ . **No g involved !**

② If f is globally Lipschitz, then (1.7) yields (1.10).

③ Examples: $V(z) := \sqrt{1 + \|z\|^2}$ for dissipative f ; or $V(z) := \sqrt{1 + \frac{1}{2}w^2 + \sigma^2(1 - \cos v)}$ for pendulum; or FHN system $V(z) = \left(1 + v^4 + Bw^2\right)^{\frac{1}{4}}$.

Lyapunov functions

Compact arguments:

$$V(y_t) - \frac{C_\lambda}{\delta} \leq e^{-\delta t} \left[V(y_0) - \frac{C_\lambda}{\delta} \right] + H(\|\mathbf{x}\|_{p\text{-var},[0,t]}), \quad \forall t \in [0, T] \quad (1.11)$$

$$H(\xi) = L_V(16C_p C_g)^p \lambda^{1-p} \xi^p + 8L_V C_p C_g \xi. \quad (1.12)$$

$$\bar{R}_\lambda(\omega) := \sum_{k=0}^{\infty} e^{-k\delta} H(\|\mathbf{x}(\theta_{-k}\omega)\|_{p\text{-var},[-1,0]}) = \sum_{k=0}^{\infty} e^{-k\delta} H(\|\mathbf{x}(\omega)\|_{p\text{-var},[-1-k,-k]}). \quad (1.13)$$

Theorem (Random pullback attractor)

$$\mathcal{A}(\omega) = \bigcap_{n=1}^{\infty} \varphi(n, \theta_{-n}\omega) \bar{B}^\varepsilon(\theta_{-n}\omega) = \lim_{n \rightarrow \infty} \varphi(n, \theta_{-n}\omega) \bar{B}^\varepsilon(\theta_{-n}\omega), \quad \forall \omega \in \Omega. \quad (1.14)$$

$$\bar{B}^\varepsilon(\omega) := V^{-1}\left(\frac{C_\lambda}{\delta} + \bar{R}_\lambda(\omega) + \varepsilon\right), \quad \forall \varepsilon \geq 0. \quad (1.15)$$

Numerical attractors for discrete systems

$$y_0^\Delta \in \mathbb{R}^d,$$

$$y_{t_{k+1}}^\Delta = y_{t_k}^\Delta + f(y_{t_k}^\Delta)\Delta + g(y_{t_k}^\Delta)x_{t_k, t_{k+1}} + Dg(y_{t_k}^\Delta)g(y_{t_k}^\Delta)\mathbb{X}_{t_k, t_{k+1}}, \quad k \in \mathbb{N}. \quad (1.16)$$

under the regular grid $\Pi = \{t_k := k\Delta\}_{k \in \mathbb{N}}$, $0 < \Delta \leq 1$.

Theorem (LHD, P. Kloeden (SINUM 2023); LHD, N.D. Cong, P. T. Hong (JDDE 2023))

Assume that $y(t, 0, y_0)$ is a solution of the rough differential equation (1.4) on $[0, T]$. Then there exists a generic constant $C = C(f, g, \|\mathbf{x}\|_{\nu, [0, T]}, T, \|y_0\|) > 0$ such that for $\Delta \ll 1$

$$\sup_{0 \leq k \leq m} \|y(t_k, 0, y_0) - y_k^\Delta\| \leq C\Delta^{3\nu-1}. \quad (1.17)$$

Assume further that f is globally Lipschitz continuous and dissipative. Then φ^h admits a numerical pullback attractor $\mathcal{A}^h \in \mathcal{L}^\rho$ which converges to the attractor $\mathcal{A}$ a.s.

$$\lim_{h \rightarrow 0} d_H(\mathcal{A}^h | \mathcal{A}) = 0 \quad \text{a.s.} \quad (1.18)$$

Compactness arguments:

$$\|y_1\|^2 \leq (\|\bar{y}_1(y_0)\| + \xi_1(\mathbf{x}, \Delta)\|y_0\|^\beta + \xi_2(\mathbf{x}, \Delta))^2 \leq \left[(1 + \varepsilon)e^{-d} + \varepsilon\right]\|y_0\|^2 + \bar{\xi}(\mathbf{x}, \Delta).$$

Fitzhugh-Nagumo neuro-system

The deterministic FHN neuron model is

$$\begin{cases} dv_t &= v_t - \frac{v_t^3}{3} - w_t + I \\ dw_t &= \varepsilon(v_t - \mu w_t + J) \end{cases} \quad (1.19)$$

f is dissipative but does not satisfy

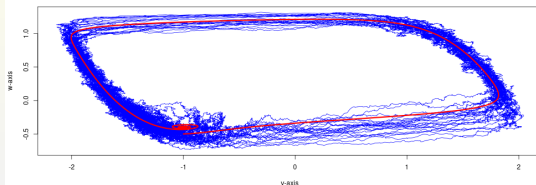
$$\left\| f(y) - \frac{\langle f(y), y \rangle}{\|y\|^2} y \right\| \leq C_f (1 + \|y\|), \quad \forall y. \quad (1.20)$$

Instead, we introduce a Lyapunov function

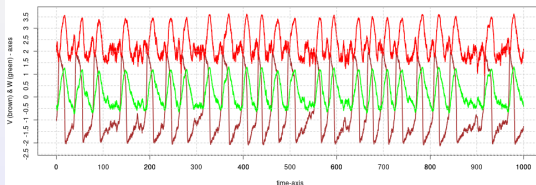
$$V(y) := \left(1 + v^4 + Bw^2\right)^{\frac{1}{4}}, \quad (1.21)$$

$$\|\nabla V\| \leq L_V. \quad (1.22)$$

Trajectory of rough FHN system. Sigma = 0.09



Time series rough FHN system



Motivations

- Stability criterion should be free of diffusion g , at most only C_g is needed.
- Data-driven approach: only f (with parameters) and realizations α are given.
- (LHD-PT Hong-ND Cong: SICON 2019): Linearization

$$d\xi = Df(\alpha_t)\xi dt + Dg(\alpha_t)\xi dx = A(t)\xi dt + C(t)\xi dx.$$

$$\xi_t = S(t, \tau, \alpha)\xi_0 + \int_{\tau}^t S(t, s, \alpha) Dg(\alpha_s)\xi_s dx_s. \quad (2.1)$$

Spectral theory works only if g is known!

- Discrete system, even Young equations: $\xi_{k,k+1} = Df(\alpha_k)\xi_k \Delta + Dg(\alpha_k)\xi_k \Delta x_k$?

Stopping time analysis

- **Continuous time:** for any fixed $\gamma \in (0, 1)$ the sequence of stopping times $\{\tau_i(\gamma, \mathbf{x}, [0, \infty))\}_{i \in \mathbb{N}}$ is defined by

$$\tau_0 = 0, \quad \tau_{i+1} := \inf \left\{ t > \tau_i : \|\mathbf{x}\|_{p\text{-var}, [\tau_i, t]} = \gamma \right\}. \quad (2.2)$$

For a fixed closed interval $I \subset [0, \infty)$, we define another sequence of stopping times $\{\tau_i^*(\gamma, \mathbf{x}, I)\}_{i \in \mathbb{N}}$ by

$$\tau_0^* = \min I, \quad \tau_{i+1}^* := \inf \left\{ t > \tau_i^* : \|\mathbf{x}\|_{p\text{-var}, [\tau_i^*, t]} = \gamma \right\} \wedge \max I. \quad (2.3)$$

Define $N^*(\gamma, \mathbf{x}, I) := \sup\{i \in \mathbb{N} : \tau_i^* < \max I\} + 1$.

- **Discrete time:** For given $\Delta > 0$, consider the sequence $t_k = k\Delta$, $k \in \mathbb{N}$. Given a fixed $\gamma > 0$, assign the starting time $\tau_0^\Delta(\gamma, \mathcal{S}, [0, \infty)) = 0$. For each $n \in \mathbb{N}$, assume $\tau_n^\Delta(\gamma, \mathcal{S}, [0, \infty)) = t_k$ is determined. Then one can define $\tau_{n+1}^\Delta(\gamma, \mathcal{S})$ by the following rule:
 - if $\sum_{\mathbf{w} \in \mathcal{S}} \mathbf{w}_{t_k, t_{k+1}}^{\beta \mathbf{w}} > \gamma$ then set $\tau_{n+1}^\Delta(\gamma, \mathcal{S}, [0, \infty)) := t_{k+1}$;
 - else set $\tau_{n+1}^\Delta(\gamma, \mathcal{S}, [0, \infty)) := \sup\{t_l > t_k : \sum_{\mathbf{w} \in \mathcal{S}} \mathbf{w}_{t_k, t_l}^{\beta \mathbf{w}} \leq \gamma\}$

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Happy 65th birthday to Thay Cong!